COMMERCIAL ITEM DESCRIPTION

RESIN, ACRYLIC, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Federal Specification U-R-00179A for Type II which is cancelled.

This Commercial Item Description covers an acrylic base denture resin formed by cold-cure (autopolymerization) suitable for denture repair.

Salient characteristics:

General. The resin shall be in accordance with ANSI/ADA Specifications No 13 unless specified otherwise herein. The resin shall be an acrylic base denture resin formed by autopolymerization suitable for use as a repair polymer. The resin polymer shall be supplied in a package containing a bottle of powder and a bottle of liquid. The resin shall be a minimum of 1 pound of powder and a minimum of liquid which is 10 percent in excess of the amount required for combining with the 1 pound of powder.

Material.

Liquid. The liquid shall be a methyl methacrylate monomer with a suitable storage polymerization inhibitor. The liquid shall be clear, colorless, and substantially free from undissolved or particulate matter when examined, without accessory magnification, against a black background and against a white background. Illumination shall be from a light which, at a point 10 inches from its source, provides an intensity of illumination of not less than 200 and not more than 350 foot candles. The liquid shall be a cross-link type monomer, which produces a cross-linked cured resin on polymerization with the powder, when processed in accordance with manufacturer's instructions.

Clarity. In addition, there shall be no visable discoloration of the liquid sample when tested as specified in test section under discoloration.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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Thermal stability. In addition, the liquid shall show no thickening when heated in the dark in a closed container at 60 ± 2°C for 24 hours.

Powder. The powder shall be polyacrylic acid esters, substituted polyacrylic acid esters, or copolymers or mixtures thereof. The powder shall be free of extraneous material such as dirt or lint.

Properties of uncured resin polymer. The uncured resin polymer, when prepared at 23.0 ± 2.0°C and as directed by the instructions accompanying the package shall be considered ready for packing when plasticity is such that the polymer separates cleanly from the walls of the glass mixing jar. This condition shall be attained in 20 minutes or less from the start of the mixing.

The resin polymer shall be capable of being intruded into a minimum of two holes of the die to a depth of 0.5 millimeter (mm) minimum when tested in accordance with the packing plasticity test of ANSI/ADA-12 for denture base polymers for Type II autopolymerized polymers.

Cured Repair resin. The resin polymer when processed according to instructions furnished by the manufacturer, shall yield a satisfactory resin repair.

Gloss The resin polymer shall have a glossy surface and shall show no surface defects, when cured in a gypsum mold lined with tin foil.

Toxicity. In addition, unless otherwise specified, prior to delivery the manufacturer shall furnish a statement directly to the Directorate of Medical Materiel, Technical Services Branch, DPSC-RST, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101-8419 stating that the repair acrylic resin does not normally irritate oral tissues and does not contain toxic ingredients in sufficient concentration to be harmful to human beings when used as directed.

Porosity and other defects. In addition, the strips of the transverse deflection test, specified herein, shall be examined for voids and other defects.

Color. In addition, the resin color shall be pink.

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Manufacturer's Instructions. In addition, the instructions shall accompany each package of resin and shall include a method for repairing the resin polymer.

Certification. The resin shall be certified by the American Dental Association. At time of award of the contract, the manufacturer shall furnish, upon request, a certificate and data showing that the brand of material to be furnished is the same in quality and brand as that appearing on the list of certified materials. Compliance with this certification requirement does not absolve the manufacturer from complete compliance with the other requirements of this specification.

Delivery schedule. Not more than two months shall have elapsed from the date of manufacture of the resin to the date of delivery to the Government.

Immediate containers. Powders shall be packaged in wide-mouth glass bottles of suitable size and design. Liquids shall be packaged in narrow-mouth, light-resistant glass bottles of suitable size and design.

Identification marking. A permanent label shall be applied to each bottle of liquid. The legend on the label shall include the manufacturer's name, item name, lot or control numbers, and the term "Flammable". The legend shall be permanent and easily legible.

Workmanship. The resin powder shall be free from foreign matter and the resin liquid shall be free of deposits, and sediment and both shall be free of any defect which detracts from its appearance or impairs its serviceability.

Unit. Package (PG). One package containing 454 grams of resin acrylic powder, and one container of liquid containing 10 percent in excess of the quantity required to combine with the powder together with instruction for use, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one package of resin, as specified. Unless otherwise specified, sampling for examination shall be conducted at inspection level S-1, AQL 1.0, and sampling for tests shall be conducted at inspection level S-1, AQL 1.0.

Examination. The resin shall be examined to determine compliance with all requirements specified herein. Resin shall be examined for defects, including but not limited to those listed below:

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DEFECTS

Liquid not free of deposits or sediment.

Powder not free of foreign matter such as dirt and lint.

Color not as specified.

Unit not complete with powder and liquid.

Instructions not included or complete.

Powder not packaged in wide-mouth glass bottles of suitable size.

Liquid not packaged in narrow-mouth, light-resistant glass bottle of suitable design.

The word FLAMMABLE not present on liquid unit label.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. There shall be two determinations per lot or batch number. Each determination shall be the average of specimens when averaging is specified in testing procedure.

Discoloration. Discoloration of the liquid shall be determined by placing 10 milliliters (ml) of the liquid into a clear glass test tube, having an inside diameter of approximately 1 cm, and by exposing this sample to bright diffuse sunlight for 24 hours. this sample shall be compared with 10 ml of the unexposed liquid placed in a similar test tube. The tubes shall be viewed perpendicularly to their long axis when backed by a white background for compliance with clarity requirements as specified herein.

Contractor certification. The contractor shall certify that the product offered meets the the salient characteristics of this description and conform to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use the covered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

Stock Number	Item Identification
6520-01-020-8270	RESIN, ACRYLIC, DENTAL, Denture Base Repair, Pink, 1 Lb.

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Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD - MB

Agent:

DLA - DM

Civil Agency Coordinating Activities:

VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2205

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53940	RESIN, ACRYLIC, DENTAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*
3b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
7c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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